12 Here are the first four terms of a quadratic sequence.

3 8 15 24

(a) Find an expression, in terms of n, for the nth term of this sequence.

n	1	2	3	4
Term	3	8	15	24
1st diff.	5	7	9	
2nd diff.	2	2		

1 mark
2nd difference 2
means n^2 coefficient 1

Subtract n^2: 2, 4, 6, 8. This is 2n.

1 mark
Remaining rule 2n

nth term = $n^2 + 2n$.

1 mark
$n^2 + 2n$

(3)

The nth term of a different sequence is $2^n + 5$

(b) Show that 36 is **not** a term of this sequence.

$2^n = 36 - 5 = 31$.

But $2^4 = 16$ and $2^5 = 32$; 31 is not a power of 2.

So 36 cannot be a term.

1 mark
31 is not a power of 2

(1)

(Total for Question 12 is 4 marks)